The University of Texas at Austin
Optimization

Homework 4 Solutions

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1. (Unit eigenvectors of stochastic matrices.) Let $P \in \mathbb{R}^{n\times n}$ have nonnegative entries and row sums equal to one. Use duality to show that $P^\mathsf{T}\lambda = \lambda$, $\lambda \geq 0$, has a nonzero solution.

   Solution: Consider the following LP and its dual:

   $$\begin{array}{rll} \max: & \mathbf{1}^\mathsf{T}\lambda & \\ \text{s.t.}: & (P^\mathsf{T} - I)\lambda = 0 & \\ & \lambda \geq 0, & \end{array} \qquad \begin{array}{rll} \min: & 0 & \\ \text{s.t.}: & (P - I)y \geq \mathbf{1}, & \\ & y \text{ free}, & \end{array}$$

   where $\mathbf{1}$ is the all-ones vector. The dual is infeasible. To see this, choose k with $y_k = \max_j y_j$. Since P is stochastic,

   $$(Py)_k = \sum_j p_{kj} y_j \leq \sum_j p_{kj} y_k = y_k.$$

   Thus $((P - I)y)_k \leq 0$, contradicting the dual constraint.

   The primal is feasible at $\lambda = 0$. If zero were its only feasible point, its optimal value would be 0, and strong duality would imply that the dual is feasible. This is a contradiction. Hence a nonzero $\lambda \geq 0$ satisfying $P^\mathsf{T}\lambda = \lambda$ exists. Dividing it by $\mathbf{1}^\mathsf{T}\lambda > 0$ gives an invariant probability distribution. □

2. (Duality in piecewise linear convex optimization.) Let

   $$v = \inf_{x\in\mathbb{R}^n} \max_{i=1,\ldots,m} (a_i^\mathsf{T} x - b_i)$$

   be finite, and let A have rows a_i^T.

   (a) Show that $-p^\mathsf{T} b \leq v$ whenever $A^\mathsf{T} p = 0$, $p \geq 0$, and $\mathbf{1}^\mathsf{T} p = 1$.

   Solution: For every x, the maximum is at least any convex combination of its entries. Therefore,

   $$\max_i (a_i^\mathsf{T} x - b_i) \geq \sum_i p_i (a_i^\mathsf{T} x - b_i) = x^\mathsf{T} A^\mathsf{T} p - p^\mathsf{T} b = -p^\mathsf{T} b.$$

   Taking the infimum over x gives $v \geq -p^\mathsf{T} b$. □

   (b) Show that the best lower bound in part (a) is equal to v.

   Solution: Introduce a scalar t. The original problem is equivalent to

   $$\begin{array}{rll} \min: & t & \\ \text{s.t.}: & Ax - t\mathbf{1} \leq b, & \qquad x, t \text{ free}. \end{array}$$

To obtain its dual, take the infimum over x, t after introducing nonnegative multipliers p:

$$\sup_{p\ge 0}\inf_{x,t}\left[t + p^{\mathsf{T}}(Ax - t\mathbf{1} - b)\right] = \sup_{p\ge 0}\inf_{x,t}\left[(A^{\mathsf{T}}p)^{\mathsf{T}}x + (1 - \mathbf{1}^{\mathsf{T}}p)t - p^{\mathsf{T}}b\right].$$

Since x and t are free, avoiding an inner value of $-\infty$ requires $A^{\mathsf{T}}p = 0$ and $\mathbf{1}^{\mathsf{T}}p = 1$. Thus the dual is

$$\begin{aligned} \max: \quad & -p^{\mathsf{T}}b \\ \text{s.t.}: \quad & A^{\mathsf{T}}p = 0, \\ & \mathbf{1}^{\mathsf{T}}p = 1, \\ & p \ge 0. \end{aligned}$$

The primal is feasible, for example at $x = 0$ and $t = \max_i(-b_i)$, and has finite value by assumption. By LP attainment and strong duality, the dual has an optimal solution with value v. □

3. Let A be symmetric. Consider

$$\begin{aligned} \min: \quad & c^{\mathsf{T}}x \\ \text{s.t.}: \quad & Ax \ge c, \qquad x \ge 0. \end{aligned}$$

Show that if $Ax^* = c$ and $x^* \ge 0$, then x^* is optimal.

Solution: The dual is

$$\begin{aligned} \max: \quad & c^{\mathsf{T}}y \\ \text{s.t.}: \quad & A^{\mathsf{T}}y \le c, \qquad y \ge 0. \end{aligned}$$

By symmetry, $y = x^*$ satisfies $A^{\mathsf{T}}y = Ax^* = c$ and $y \ge 0$. Thus x^* is primal feasible, $y = x^*$ is dual feasible, and their objective values are equal. Weak duality proves that x^* is optimal. □

4. (Robust linear programming.) Let

$$\mathcal{U} = \{b \in \mathbb{R}^m : 0 \le b_i \le 1,\ i = 1, \dots, m,\ \textstyle\sum_i b_i \le 1\}.$$

Show that minimizing $c_1x_1 + c_2x_2 + c_3x_3$ subject to $Ax \ge b$ for every $b \in \mathcal{U}$ is equivalent to imposing $Ax \ge \mathbf{1}$.

Solution: For each i, the unit vector e_i belongs to $\mathcal{U}$. Hence the robust constraint implies $(Ax)_i \ge 1$ for each i, or $Ax \ge \mathbf{1}$. Conversely, every $b \in \mathcal{U}$ satisfies $b \le \mathbf{1}$, so $Ax \ge \mathbf{1}$ implies $Ax \ge b$ for every $b \in \mathcal{U}$. The feasible sets coincide, and the objective is unchanged. □

5. (Robust linear programming — optional.) Give a finite formulation of

$$\begin{aligned} \min: \quad & c^{\mathsf{T}}x \\ \text{s.t.}: \quad & Ax \le b, \qquad \forall b \in \mathcal{U}_1, \\ & a_2^{\mathsf{T}}x \le b_2, \qquad \forall a_2 \in \mathcal{U}_2, \\ & a_3^{\mathsf{T}}x \le b_3, \qquad \forall a_3 \in \mathcal{U}_3, \end{aligned}$$

where $\mathcal{U}_1 \subseteq \mathbb{R}^m$ is a polytope and

$$\mathcal{U}_2 = \{a : D_2a = d_2,\ a \ge 0\}, \qquad \mathcal{U}_3 = \{a : D_3a \ge d_3,\ a \le 0\}.$$

Here $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$, and $D_j \in \mathbb{R}^{r_j \times n}$. We use Ax in the first constraint to match the stated dimensions, and label the last set $\mathcal{U}_3$, correcting the two notation inconsistencies in the handout.

Solution: Assume first that the uncertainty sets are nonempty. For the first constraint, define

$$\underline{b}_i = \min_{b\in\mathcal{U}_1} b_i, \qquad i = 1, \dots, m.$$

These are finite constants, each obtained by solving an LP. The first robust constraint is equivalent to $Ax \le \underline{b}$.

For the second constraint, hold x fixed and maximize over a. The inner LP and its dual are

$$\begin{array}{rll} \max: & x^\mathsf{T} a & \\ \text{s.t.}: & D_2 a = d_2 & \\ & a \ge 0, & \end{array} \qquad\qquad \begin{array}{rll} \min: & d_2^\mathsf{T} y_2 & \\ \text{s.t.}: & D_2^\mathsf{T} y_2 \ge x, & \\ & y_2 \text{ free.} & \end{array}$$

Thus the robust constraint is equivalent to the existence of y_2 such that $D_2^\mathsf{T} y_2 \ge x$ and $d_2^\mathsf{T} y_2 \le b_2$.

For the third constraint, set $z = -a \ge 0$. Its inner LP and dual become

$$\begin{array}{rll} \max: & (-x)^\mathsf{T} z & \\ \text{s.t.}: & D_3 z \le -d_3 & \\ & z \ge 0, & \end{array} \qquad\qquad \begin{array}{rll} \min: & -d_3^\mathsf{T} \mu_3 & \\ \text{s.t.}: & D_3^\mathsf{T} \mu_3 \ge -x, & \\ & \mu_3 \ge 0. & \end{array}$$

Therefore the third robust constraint is equivalent to the existence of $\mu_3 \ge 0$ with $D_3^\mathsf{T}\mu_3 \ge -x$ and $-d_3^\mathsf{T}\mu_3 \le b_3$. In both cases, weak duality proves that the certificate suffices; if robustness holds, the nonempty inner LP is bounded above, and strong duality supplies such a certificate.

Combining the three blocks gives the finite LP

$$\begin{array}{rll} \min: & c^\mathsf{T} x & \\ \text{s.t.}: & Ax \le \underline{b}, & \\ & D_2^\mathsf{T} y_2 \ge x, \qquad d_2^\mathsf{T} y_2 \le b_2, & \\ & D_3^\mathsf{T} \mu_3 \ge -x, \qquad -d_3^\mathsf{T} \mu_3 \le b_3, & \\ & \mu_3 \ge 0, \qquad x \in \mathbb{R}^n, \quad y_2 \in \mathbb{R}^{r_2}. & \end{array}$$

If an uncertainty set is empty, its universal constraint is vacuous, so omit its block. The sets $\mathcal{U}_2$ and $\mathcal{U}_3$ may be unbounded: a decision x with an unbounded inner objective is excluded by the corresponding dual-feasibility conditions. □